GCE A LEVEL MARKING SCHEME

PREDICTED – ADVANCED

A LEVEL
ECONOMICS – COMPONENT 1
A520U10-1 (Predicted – Advanced)

About this marking scheme

The purpose of this marking scheme is to provide teachers, learners and other interested parties with an understanding of the assessment criteria used to assess this specific assessment.

A list of indicative content suggests the range of economic concepts, theory, issues and arguments which might be included in learners' answers. It is not intended to be exhaustive and learners do not have to include all of the indicative content to reach the highest level of the mark scheme.

The level-based mark schemes sub-divide the total mark to allocate to individual assessment objectives. A mark will be awarded for each assessment objective targeted in the question, then totalled to give an overall mark for the question.

This Advanced predicted paper is pitched above A-Level baseline in places, incorporating material that approaches first-year undergraduate economics (e.g. Cournot oligopoly, Solow growth, Lucas critique, Weitzman prices-vs-quantities). Marking should nevertheless follow the standard A-Level banded rubric – credit should be given for rigorous economic reasoning, accurate use of data, and supported judgements.

GENERAL MARKING GUIDANCE

Positive Marking

It should be remembered that learners are writing under examination conditions and credit should be given for what the learner writes, rather than adopting the approach of penalising for any omissions.

Banded Mark Schemes

Where a banded mark scheme is used, the mark awarded should reflect the overall quality of the response against the descriptors for each Assessment Objective. Examiners should consider the response as a whole rather than applying a purely mechanistic count of points.

SECTION A

Q	Answer	AO
1	**B**	AO2
2	**B**	AO2
3	**B**	AO2
4	**B**	AO1
5	**C**	AO2
6	**B**	AO1
7	**B**	AO2
8	**B**	AO1
9	**A**	AO1
10	**B**	AO1
11	**C**	AO1
12	**A**	AO1
13	**B**	AO1
14	**B**	AO2
15	**C**	AO1
16	**B**	AO1
17	**B**	AO1
18	**B**	AO2
19	**C**	AO2
20	**B**	AO1

AO1: 12

AO2: 8

Examiner's notes for numerical questions

Q1 (B):

$TC = 200 + 30Q - 2Q^2 + 0.2Q^3$. FC = 200 (the constant term). AFC at Q=10 is 200/10 = £20. $MC = dTC/dQ = 30 - 4Q + 0.6Q^2$. At Q = 10: MC = 30 − 40 + 60 = £50.

Q2 (B):

Cournot symmetric Nash equilibrium with linear demand $P = a - bQ$ and constant MC = c: $q^* = (a - c) / (3b)$. With a = 120, b = 1, c = 30: $q^* = 90/3 = 30$ per firm. Industry output is 60, price is 60.

Q3 (B):

Keynesian multiplier with proportional tax and imports: $k = 1 / (1 - MPC(1 - t) + MPM) = 1 / (1 - 0.8 \times 0.75 + 0.15) = 1 / (1 - 0.6 + 0.15) = 1/0.55 \approx 1.818$.

Q5 (C):

Market 1: $MR_1 = 100 - 2Q_1 = 20 \rightarrow Q_1 = 40$, $P_1 = £60$. Market 2: $MR_2 = 80 - 4Q_2 = 20 \rightarrow Q_2 = 15$, $P_2 = £50$. (Higher price in the more inelastic market, as predicted by third-degree price discrimination.)

Q19 (C):

$P = 200 - 2Q \rightarrow TR = 200Q - 2Q^2$, $MR = 200 - 4Q$. $TC = 50 + 20Q + Q^2 \rightarrow MC = 20 + 2Q$. Set MR = MC: $200 - 4Q = 20 + 2Q \rightarrow Q^* = 30$, $P^* = 140$. TR = 4,200; TC = 50 + 600 + 900 = 1,550. Profit = £2,650.

SECTION B

21. Using the data, explain the relationship shown between the US 10Y–2Y Treasury yield spread and GDP growth. [4]

Band	AO1 (1 mark)	AO2 (1 mark)	AO3 (2 marks)
2			**2 marks** **Good analysis** Developed chain from yield-curve inversion to recession via expectations and monetary policy.
1	1 mark **Limited knowledge** Understanding of the yield curve and/or the term premium/expectations hypothesis.	1 mark **Limited application** Specific reference to the data points (e.g. inversion in 2019/2022, subsequent growth decline).	1 mark **Limited analysis** Chain of reasoning linking the yield spread to GDP but underdeveloped.
0	**0 marks**	**0 marks**	**0 marks**

Indicative content

AO1

The yield spread (10-year minus 2-year Treasury yield) is a widely used leading indicator. Under the expectations hypothesis, the long-term yield reflects expectations of future short-term rates; an inversion signals that investors expect central bank rate cuts, typically because they anticipate a recession. The term premium normally makes longer yields higher than shorter ones, so inversion is unusual and noteworthy.

AO2

The data shows the 10Y–2Y spread turning negative (inverted) in 2019 and again from mid-2022 through 2024. GDP growth fell sharply in 2020 (the Covid recession) – consistent with a recession signal from the 2019 inversion – and slowed in 2023–24 to below 2%, again following the 2022 inversion. The relationship is not perfectly contemporaneous: inversions typically lead recessions by 6–24 months.

AO3

Chain: markets price in future monetary loosening → long-dated yields fall below short-dated yields (inversion) → this both signals and contributes to a slowdown because it compresses bank net interest margins (banks borrow short, lend long) → credit supply tightens → investment and consumption slow → GDP growth falls. Every US recession since the 1960s has been preceded by an inversion, though the 2022 inversion is noted for its unusual length without (yet) producing a recession – possibly due to post-pandemic fiscal support and resilient labour markets.

Allow other relevant points.

22. Discuss whether tradable pollution permits are superior to a Pigouvian (emissions) tax when the marginal abatement cost curve is uncertain. [7]

Band	AO2 (2 marks)	AO3 (2 marks)	AO4 (3 marks)
3			**3 marks** **Excellent evaluation** Clear supported judgement on the conditions under which permits dominate tax (or vice versa), engaging with the relative slopes of MAC and MD.
2	**2 marks** **Good application** Effective use of diagram/example to show uncertainty effects.	**2 marks** **Good analysis** Developed chain of reasoning linking MAC uncertainty to welfare loss under each instrument.	**2 marks** **Good evaluation** Developed counterarguments covering both sides (quantity certainty vs price certainty).
1	1 mark **Limited application** Diagram/example referenced superficially.	1 mark **Limited analysis** Chain of reasoning underdeveloped.	1 mark **Limited evaluation** Counterpoint identified but not developed.
0	**0 marks**	**0 marks**	**0 marks**

Indicative content

AO2

Both instruments can achieve the same emissions target if marginal abatement cost (MAC) is known with certainty: set the tax rate equal to marginal damage (MD) at the optimal emissions level, or issue that many permits and let the market price converge to the same value. The question is what happens when MAC is uncertain. A Pigouvian tax fixes the price of emissions; the quantity abated adjusts to whatever level firms choose at that price – so if MAC turns out higher than expected, firms abate less than planned. Tradable permits fix the quantity of emissions; the permit price adjusts – so if MAC is higher than expected, the permit price rises.

AO3

Weitzman (1974) 'Prices vs Quantities': when MAC is uncertain, the welfare loss under each instrument depends on the relative slopes of MAC and MD. If MD is steep (each extra tonne of pollution causes rapidly rising damage – e.g. localised toxic emissions, or approaching a climate tipping point), the quantity instrument (permits) is preferred because quantity certainty matters more than price certainty. If MD is flat relative to MAC (each tonne causes roughly the same marginal damage, as approximately true for GHG emissions over short horizons), the price instrument (tax) is preferred because it avoids large cost spikes from a binding permit cap when MAC turns out high.

AO4

Counterarguments and qualifications: (i) In practice, real-world schemes are hybrids. The EU ETS has been reformed to include a Market Stability Reserve and price floors/ceilings, making it closer to a price instrument in some conditions – so the stark Weitzman dichotomy is softened. (ii) Permits create opportunities for rent extraction (if freely allocated, as in early EU ETS phases, incumbents receive windfall profits). (iii) Taxes are simpler administratively and generate certain revenue streams for government, useful for green investment or to offset regressive impacts. (iv) Permits can be more politically palatable because the visible cost is on industry rather than consumers, even though the incidence is ultimately similar. (v) For climate change specifically, most economists have argued for carbon pricing via a tax precisely because MD (discounted global damages) is relatively flat over the relevant range while MAC is highly uncertain – though opinions differ on how urgent cumulative-emissions constraints make quantity certainty. (vi) Practical considerations such as monitoring costs, verification of emissions, leakage and competitiveness effects apply similarly under both instruments. Overall: neither dominates unconditionally – the superior instrument depends on the relative slopes of MAC and MD, administrative feasibility, and political economy.

Allow other relevant points.

23. Using the payoff matrix, explain the Nash equilibrium in this market and assess whether the incumbent's threat to fight is credible. [4]

Band	AO1 (1 mark)	AO2 (2 marks)	AO3 (1 mark)
2		**2 marks** **Good application** Matrix used to identify the subgame-perfect Nash equilibrium and to show that the incumbent's 'fight' payoff is lower than 'accommodate' post-entry.	
1	1 mark **Limited knowledge** Understanding of Nash equilibrium and/or credibility of threats.	1 mark **Limited application** The matrix data is used to identify the equilibrium.	1 mark **Limited analysis** Chain of reasoning about why the fight threat fails.
0	**0 marks**	**0 marks**	**0 marks**

Indicative content

AO1

A Nash equilibrium is a profile of strategies such that no player can improve by deviating unilaterally. A threat is credible only if the threatening party would actually find it optimal to carry out the threat after the other player acts – in other words, only if the strategy is part of a subgame-perfect equilibrium.

AO2 / AO3

Working through the matrix: if Firm E stays out, Firm I earns £100m (monopoly) regardless of its announced strategy, and Firm E earns £0. If Firm E enters, Firm I compares fighting (£30m) with accommodating (£60m) and chooses Accommodate because 60 > 30. Firm E, anticipating that Fight will not be chosen, compares Enter (£40m) with Stay Out (£0) and chooses Enter. Subgame-perfect Nash equilibrium: (Enter, Accommodate). The threat to fight is non-credible because once entry has occurred, fighting is ex-post suboptimal for the incumbent. To make the threat credible, the incumbent would need a commitment device – e.g. sunk investment in excess capacity, a reputation-building strategy across multiple markets, or contractual penalties – that would raise the payoff to fighting or lower the payoff to accommodating. This is the Selten critique of announced threats and the foundation of strategic entry deterrence theory.

Allow other relevant points.

24. Using the data, discuss whether supply-side reforms are likely to resolve the UK's productivity stagnation. [8]

Band	AO2 (2 marks)	AO3 (3 marks)	AO4 (3 marks)
3		**3 marks** **Excellent analysis** Developed chains linking specific reforms to productivity through endogenous/exogenous growth channels.	**3 marks** **Excellent evaluation** Well-reasoned supported judgement, engaging with time lags, magnitude and uncertainty.
2	**2 marks** **Good application** Data used effectively (the 1% cumulative gain vs a normal trend would give ~15%+ over a decade).	**2 marks** **Good analysis** Developed chain of reasoning about the mechanism by which supply-side reforms lift productivity.	**2 marks** **Good evaluation** Developed counterargument about the limits of supply-side reforms.
1	1 mark **Limited application.**	1 mark **Limited analysis.**	1 mark **Limited evaluation.**
0	**0 marks**	**0 marks**	**0 marks**

Indicative content

AO2

The data shows UK output per hour rising from 100.0 in 2015 to only 101.5 in 2024 – approximately 0.15% per year, far below the pre-financial crisis UK trend of 2.0–2.3% per year and below G7 peers. Cumulative productivity gain of only ~1.5% over nine years represents a very large shortfall versus the counterfactual trend (~20%). The question focuses on whether supply-side reforms (investment in R&D, infrastructure, skills; planning reform; tax reform; competition policy; labour market reforms) can close this gap.

AO3

Mechanism 1: investment in physical and human capital directly raises the capital-labour ratio and the stock of skills, shifting the LRAS / production function upward. Infrastructure investment (transport, digital) reduces transaction costs for the whole economy. Mechanism 2: endogenous growth theory (Romer, Aghion-Howitt): R&D tax credits, skilled migration, university-industry links raise the rate of innovation and total factor productivity. Mechanism 3: planning reform could unlock housing supply in high-productivity cities (agglomeration effects – London and the South East are more productive), addressing 'missing middle' clustering. Mechanism 4: competition policy and removal of trade barriers raise allocative and dynamic efficiency. Mechanism 5: skills reforms (apprenticeships, adult learning, FE investment) raise human capital and reduce the productivity gap between top and laggard firms.

AO4

Counterarguments and evaluation: (i) Time lags – most supply-side policies take 5–15 years to raise measured productivity. The 2024–25 government cannot expect quick wins. (ii) The UK's productivity puzzle is partly cyclical / hysteresis-driven (GFC, austerity, Brexit, pandemic) and partly structural; supply-side reforms may not fix a problem rooted in weak aggregate demand, underinvestment and uncertainty. (iii) Measurement issues – services-heavy economies suffer from productivity measurement challenges. (iv) Brexit effects on trade frictions, skilled migration and FDI may offset supply-side reform gains. (v) Opportunity cost and fiscal constraints – R&D and infrastructure investment compete with other spending in a tight fiscal environment (high debt, high interest rates). (vi) International evidence is mixed – supply-side reforms have worked well in some contexts (Germany's Hartz reforms, Ireland's FDI and education strategy) but poorly in others. (vii) Distributional concerns – many supply-side reforms (e.g. planning liberalisation, skills reform) face political resistance. (viii) Complementary demand management matters: if reforms raise capacity but demand remains weak, measured productivity may not rise (the Verdoorn effect works both ways). Overall judgement: supply-side reforms are necessary but not sufficient; they must be well-designed, sustained for a decade or more, and combined with adequate demand management to close the productivity gap. Meaningful improvement is plausible but gradual and uncertain.

Allow other relevant points.

25. Using the data, discuss whether Sri Lanka is more developed than South Africa. [8]

Band	AO2 (2 marks)	AO3 (3 marks)	AO4 (3 marks)
3		**3 marks** **Excellent analysis** Well-developed chain using multiple indicators (HDI, IHDI, Gini, MPI) to argue both sides.	**3 marks** **Excellent evaluation** Supported judgement recognising the multidimensional nature of development and data limitations.
2	**2 marks** **Good application** Data used effectively on both sides of the comparison.	**2 marks** **Good analysis** Developed chain about why Sri Lanka scores higher on most indicators.	**2 marks** **Good evaluation** Developed counterargument about South Africa's advantages or Sri Lanka's recent setbacks.
1	1 mark **Limited application.**	1 mark **Limited analysis.**	1 mark **Limited evaluation.**
0	**0 marks**	**0 marks**	**0 marks**

Indicative content

AO2

Indicators suggest Sri Lanka is more developed:

HDI higher (0.782 vs 0.717) – Sri Lanka in High HD, South Africa also High but lower. Life expectancy higher (76.4 vs 65.3 – a striking 11-year gap). Much lower Gini (37.7 vs 63.0 – one of the highest in the world for South Africa). MPI dramatically lower (0.009 vs 0.025) – fewer people in multidimensional poverty. Much higher IHDI (0.697 vs 0.478) – the inequality discount narrows the raw HDI gap and then reverses it strongly.

Indicators suggest South Africa may be more developed:

GNI per capita (PPP$) slightly higher ($13,552 vs $13,090). Mean years of schooling higher (11.4 vs 10.8). Stronger democratic institutions and more diversified economy.

AO3

Development is multidimensional – income alone is a poor measure; Amartya Sen's capabilities approach emphasises real freedoms to achieve valuable outcomes. The HDI aggregates health, education and income; the IHDI additionally accounts for inequality. South Africa's extreme inequality (Gini 63, highest or near-highest in the world) means the average figures mask very unequal outcomes – roughly 30% of IHDI discount vs ~11% for Sri Lanka. The MPI captures overlapping deprivations in health, education and living standards – Sri Lanka performs much better because of historically strong public provision of health and education.

AO4

Counterarguments and qualifications: (i) Timing – the data is pre-2022 economic crisis in Sri Lanka, which saw sovereign default, shortages of fuel and medicines, IMF programme, and sharp drops in real income. Post-crisis figures would likely narrow or reverse some gaps. (ii) Gender – on the Gender Inequality Index, South Africa scores better than Sri Lanka on economic participation though worse on other dimensions; this is a dimension both indices partly capture. (iii) Environmental sustainability – not captured by HDI/IHDI; South Africa is carbon-intensive but also a regional environmental leader in some areas. (iv) Freedoms – political, civil, press freedom – South

Africa scores well on democracy and press freedom despite state capture concerns; Sri Lanka has faced civil war legacies and authoritarian episodes. (v) Infrastructure and services: South Africa has more advanced financial markets, logistics, manufacturing base – potentially relevant for capability expansion. (vi) Happy Planet Index, Social Progress Index, World Happiness Report would provide fuller pictures. Overall judgement: on the indicators shown – HDI, IHDI, Gini, MPI – Sri Lanka is more developed in most dimensions, and the IHDI gap is particularly revealing. But the 2022 crisis, the absence of environmental and freedom dimensions, and South Africa's institutional strengths mean a more tentative conclusion is warranted. The comparison strongly illustrates the limitations of single-indicator development comparisons.

Allow other relevant points.

26. Using supply and demand diagram(s), discuss whether rising mortgage rates or structural supply constraints are more important in explaining recent UK house price movements. [9]

Band	AO1 (2 marks)	AO2 (2 marks)	AO3 / AO4
3			**3 marks (AO4)** **Excellent supported judgement balancing cyclical (rates) vs structural (supply) factors over short and long runs.**
2	**2 marks** Accurate S&D diagrams for both a demand-side and a supply-side explanation.	**2 marks** Effective use of data (mortgage rate rise from 1.2% to 5.4%, housebuilding below target, price fall of 1.8% in 2023 then rise of 2.4% in 2024).	**2 marks (AO3)** **Developed analysis of one mechanism. 2 marks (AO4)** Developed counterargument considering the alternative driver.
1	1 mark Diagram present but inaccurate, or only one side.	1 mark Data used on one side only or superficially.	1 mark (AO3) / 1 mark (AO4) Chain of reasoning underdeveloped.
0	**0 marks**	**0 marks**	**0 marks**

Indicative content

AO1

Diagram 1 (cyclical – rates channel): In the market for houses, a rise in mortgage rates reduces the affordability of borrowing, shifting the demand curve leftward from D_1 to D_2. Given short-run price-inelastic supply (slow construction, planning delays), price falls substantially while quantity transacted falls only a little. Diagram 2 (structural – supply shortage): Over a longer horizon, population and household formation growth shifts demand rightward from D_1 to D_2, while rigid supply (planning constraints, land-banking, slow response of housebuilding) keeps the supply curve nearly vertical, so the same rightward demand shift causes a large price rise with little quantity increase.

AO2

The data shows: 2-year fixed mortgage rates rose from 1.2% in 2021 to 5.4% by end-2023, falling slightly to 4.8% in 2024. Average UK house prices fell 1.8% in 2023 (consistent with the rates channel), then rose 2.4% in 2024 even as rates remained high (consistent with structural pressure). Housebuilding completions of ~170,000 per year in 2023 remained well below the government target of 300,000, and net household formation continued at ~200,000 per year. Real-terms house prices have roughly trebled relative to earnings since the 1990s.

AO3

Developed chain (cyclical, rates): Bank Rate rises from 0.1% to 5.25% → mortgage rates rise sharply → monthly mortgage payment for a given loan size rises (e.g. a £250k 25-year mortgage at 1.2% ≈ £963/month; at 5.4% ≈ £1,515/month, a 57% increase) → affordability constraints tighten, fewer buyers can meet lender stress tests → demand shifts left → nominal prices fall modestly, real prices fall more, transactions collapse (especially first-time buyers and re-mortgagors). This explains the 2023 price fall neatly. Developed chain (structural, supply): UK housebuilding has averaged 150,000–180,000 per year versus a household formation rate of 200,000+; net shortfall accumulates. Planning system (Town and Country Planning Act 1947) imposes restrictive green belt and local authority discretion – PES of housing is very low in the SR and LR. Demand continues to grow from population, immigration and smaller households. This explains why prices are high in absolute terms and why they rebounded in 2024 despite high rates.

AO4

Evaluation: (i) Time horizon matters. In the short run, interest rate shocks dominate the data – the 4.2pp rate rise in two years is a very large demand shock and produced the 2023 price dip. In the long run, structural supply constraints dominate – the trebling of real house prices since 1996 cannot be explained by rates, which have fallen over that period. (ii) Market segmentation – cash buyers, investors and equity-rich buyers are less rate-sensitive; first-time buyers most affected. The mix matters for the aggregate response. (iii) Regional variation – London is more rate-sensitive but also more supply-constrained; the North shows different dynamics. (iv) Government interventions – Help to Buy, stamp duty changes, shared ownership schemes – all shift demand and obscure the underlying drivers. (v) Expectations and speculation – housing is a store of value; expectations of future price rises feed current demand, amplifying cyclical movements. (vi) Mortgage market structure – most UK mortgages are 2–5 year fixes, so rate pass-through is gradual as people roll off old fixes, delaying and smoothing the cyclical response. (vii) Supply-side reforms (planning) have multi-year lags – even effective reform today would not change the 2024 market. Overall judgement: mortgage rates dominate short-run movements (and explain the 2023 dip); structural supply constraints dominate long-run levels and explain why prices rebounded in 2024 and why UK house prices are high relative to incomes by international comparison. A complete explanation requires both, weighted by the horizon.

Allow other relevant points.

AO Grid

	AO1	AO2	AO3	AO4	Total
Q1–20	**12**	**8**			**20**
Q21	1	1	2		4
Q22		2	2	3	7
Q23	1	2	1		4
Q24		2	3	3	8
Q25		2	3	3	8
Q26	2	2	2	3	9
Total	**16**	**19**	**13**	**12**	**60**

Note: AO mix for the Advanced paper is slightly weighted toward AO1 in Section A reflecting the more technical/theoretical nature of several MCQs (Cournot, Solow, Lucas critique, Dutch disease, etc.). Total marks remain 60.